

Synthetic Metacognition for Managing Tactical Complexity

Dr. Bonnie Johnson, Naval Postgraduate School, Sandy Spring, MD, 20860, USA

The integration of artificial intelligence (AI) into military operations has revolutionized the landscape of tactical decision-making, offering enhanced efficiency, accuracy, and adaptability. However, for AI to become an effective partner in complex, high-stakes environments such as military operations, it is essential that it possesses the ability to self-monitor, assess, and adjust its decision-making processes. This paper explores the concept of synthetic metacognition in the context of military tactical decision-making, proposing a framework in which AI systems can evaluate and improve their decision-making strategies in real-time. By drawing from cognitive science and machine learning principles, the paper introduces metacognitive concepts for AI to not only make decisions but also reflect on its reasoning, identify potential biases or errors, and adapt strategies to optimize mission outcomes. The paper examines how these metacognitive capabilities could be integrated into autonomous systems, decision support tools, and human-AI collaborative environments, ensuring that AI systems contribute to more robust, resilient, and flexible military operations. A longer extended paper will describe tactical scenarios for which metacognitive AI can play a beneficial role. The extended paper will discuss the challenges of implementing AI metacognition in complex, dynamic, and unpredictable battlefields, highlighting the ethical, technical, and operational considerations for future military AI applications.

Tactical military environments are becoming increasingly complex.¹ Military conflicts are inherently complex in terms of dynamic and rapidly changing adversarial events and tempos, known and unknown threat objects, uncertainty in situation awareness, many varied courses of action, and critical decisions with severe consequences.² Technological advances in warfare systems (i.e., autonomous systems, artificial intelligence (AI), directed energy weapons, cyber-attacks) have added to this “fog of war.”³ Decision-makers must make sense of their complex battlespace surroundings and must decide how best to accomplish missions for successful tactical outcomes.

The use of AI systems by autonomous systems, battle management decision aids, and warfighter-AI teams, offers solutions for improving tactical decision-

making. Autonomous systems, equipped with advanced sensors and algorithms, can process large volumes of real-time data to make independent decisions on maneuvers, resource allocation, and threat mitigation, enhancing operational speed and precision.⁴ Battle management decision aids can provide commanders with actionable insights through visualizations, predictive analytics, and scenario modeling, enabling informed decision-making in complex and fast-paced environments.⁵ When paired with warfighter-AI teams, these technologies foster a collaborative approach where AI systems augment human expertise by offering recommendations, highlighting potential risks, and supporting tactical planning, while warfighters provide the critical context and judgment needed for nuanced decision-making.⁶ This synergy of machine and human

intelligence supports a dynamic, adaptive, and resilient decision-making process that can respond effectively to evolving battlefield conditions.

The use of AI in tactical decision-making, while offering significant advantages in terms of speed and accuracy, also introduces a range of risks and safety concerns.⁷ One of the primary risks is the potential for AI systems to make incorrect or biased decisions, particularly in complex and unpredictable environments like the battlefield. This can occur due to faulty algorithms, inadequate training data, or an inability to properly interpret ambiguous or incomplete information. Such errors could lead to unintended escalation of conflict, misidentification of targets, or even friendly fire incidents, resulting in severe consequences.

Another concern is the lack of transparency in AI decision-making processes.⁸ Many AI systems, particularly those based on deep learning, operate as "black boxes," where even developers may not fully understand how the system reaches its conclusions. This opacity can be dangerous in high-stakes scenarios, as it makes it difficult to trust or verify AI decisions, especially in situations requiring immediate and critical judgments. Additionally, the possibility of adversaries exploiting vulnerabilities in AI systems—such as through cyberattacks or adversarial machine learning—raises concerns about the security and integrity of AI-driven decision-making.

Finally, the delegation of life-and-death decisions to AI systems poses ethical dilemmas, particularly regarding accountability and responsibility.⁹ If an autonomous system makes a catastrophic error, it may be unclear who is responsible—the developers, the commanders, or the machine itself. This ambiguity could complicate legal and moral frameworks in military operations, especially in cases of civilian harm or violations of international law.

This short preliminary paper aims to introduce the concept of synthetic metacognition in AI, providing a definition and exploring its potential to enhance decision making in the face of tactical complexity. The paper discusses key concepts related to how metacognitive capabilities can allow AI to reflect on its decision-making processes, assess its reasoning, and adapt its strategies in real time, thus increasing transparency and reliability in high-stakes environments. The longer version of this paper will delve deeper into current research trends in metacognition, outlining the specific metacognitive functionalities that can be integrated into AI systems. Additionally, it will examine tactical scenarios in which metacognition can improve AI-driven decision-making, offering insights into how these enhanced capabilities

could support more effective, flexible, and trustworthy military operations.

SYNTHETIC METACOGNITION

Metacognition, or "knowledge of cognition" refers to what a system knows about itself and about its own cognition.¹⁰ Figure 1 shows the process of thinking as a cycle of cognition and metacognition. "Synthetic" metacognition is the ability to embed intelligent engineered systems (or machine agents) with capabilities to have self-awareness and heightened awareness of their surroundings, situations, and other human and machine agents they may interact with. Synthetic metacognition is an engineering strategy that enables intelligent agents to identify complex situations and potential failures, and to enable self-healing and self-management for safe and desired behavior.¹¹

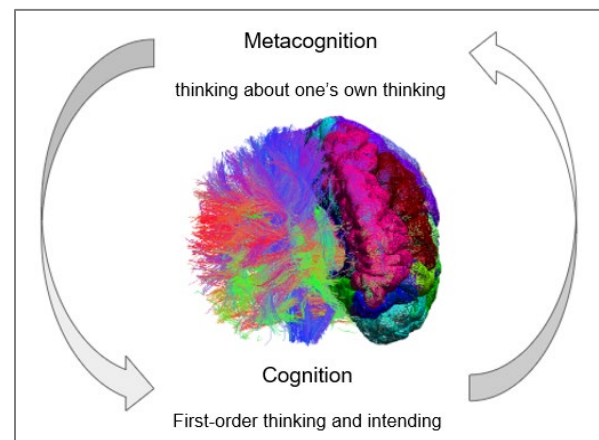


FIGURE 1. A continuous cycle of first-order thinking and intending (cognition) and more deeply thinking about one's own thinking (metacognition)¹²

Synthetic metacognition (or metacognitive AI) describes a concept for a future embedded "self-awareness" capability within intelligent machine agents. The concept is for future metacognitive machine agents to have a heightened understanding of themselves, their situation, and their "teammates" (humans and other machine agents) through new abilities of self-monitoring, self-assessment, goal-setting, self-control, and reasoning (shown in Figure 2). These capabilities enable machine agents to "manage complexity" – to determine when they have encountered a complex situation, to evaluate uncertainty, to detect anomalies, to identify new or unfamiliar situations, to evaluate complexity in the situation and in themselves, to identify and predict high-risk courses of action, to identify and predict undesirable

emergent behavior, and to establish effective goals and adapt to complex situations.

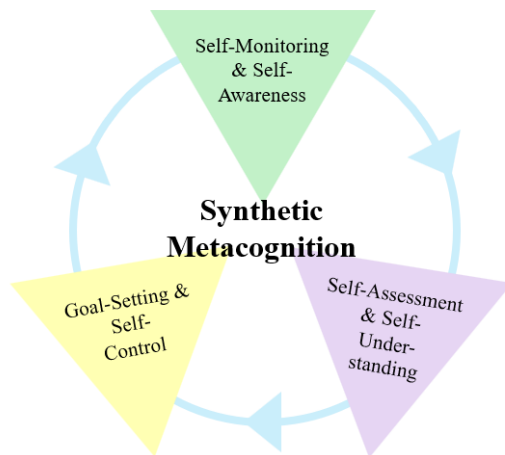


FIGURE 2. Three cornerstones of synthetic metacognitive capabilities

Research in the field of synthetic metacognition is in its early stages but is growing rapidly as it holds the potential to significantly enhance AI systems, making them more adaptive, efficient, and capable of self-improvement. The concept of developing metacognition for artificial intelligent systems originated as a series of theoretical papers written a decade ago.^{13,14,15} More recently, the Computer Science community has embraced the concepts and is making progress on various aspects of metacognitive capabilities within AI systems.^{16,17} While synthetic metacognition is a relatively new concept, computer scientists are developing frameworks such as the Carina metacognitive architecture,¹⁸ the Transparency, Reasoning, Adaptation, and Perception (TRAP) approach,¹⁹ and decision-driven AI self-assessment.²⁰

EXPLAINABILITY AND TRUST

Explainability and trust are central features of decision-making involving AI systems. Recent advances in AI decision aids have focused on methods to ensure humans relying on automated decision aids have insight into how AI decisions are formed to ensure appropriate trust in AI recommendations. AI Explainability (XAI) and human-machine trust are particularly important in tactical decision-making.

Synthetic metacognition introduces the novel capability of reverse XAI and trust. In this concept, the AI machine agents would continuously assess their human partners – to understand the human commands and behaviors. The AI agents would determine the “trust-

worthiness” of their human partners. This feature enables two-way explainability and two-way trust between humans and AI systems, as illustrated in Figure 3. This opens avenues of research into new possibilities for more symbiotic collaboration between humans and machines for complex decision-making. Metacognition embedded in the AI system overcomes the traditional limitations of humans controlling the relationship. New forms of collaboration are possible that promote more effective use of both human and AI cognitive skills.



FIGURE 3. Two-way explainability and trust between warfighters and metacognitive AI decision aids

Synthetic metacognition introduces the novel concept of machine-to-machine XAI and trust. This concept applies to situations involving multiple AI agents, such as autonomous tactical systems. The concept envisions synthetic metacognitive autonomous systems that assess themselves and each other, as illustrated in Figure 4. Multiple metacognitive agents will require machine-to-machine XAI and trust to self-organize, decide to collaborate and learn, develop goals for desired emergent behavior, and identify and curtail undesired individual and emergent behavior.

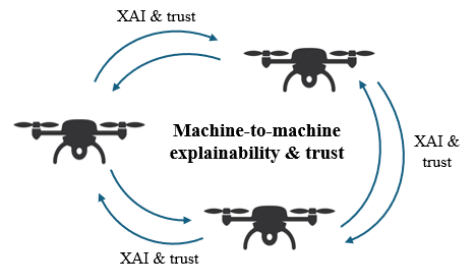


FIGURE 4. Metacognitive drones with machine-to-machine explainability and trust

Embedding metacognition into AI systems fosters two-way transparency and clarity, making AI decisions more understandable to human operators and enabling AI systems to assess human behavior and commands. Likewise, when multiple AI systems can evaluate and communicate their strategies with each other using

machine-to-machine metacognitive XAI and trust capabilities, it facilitates smoother cooperation, reducing the risk of conflicting or inefficient actions. These qualities enable better management of tactical complexity, as humans and AI systems, or groups of autonomous systems, team up and collaboratively contribute their cognitive strengths. In such environments, trust and explainability are not just beneficial; they are essential for reliable, adaptable partnerships and teams in high-stakes decision-making.

CONCLUSION

This short paper will be expanded into a longer extended version which will describe synthetic metacognitive capabilities in greater detail. The extended version will provide examples of tactical scenarios in which metacognitive AI systems can enhance the management of tactical complexity as a decision aid team with human warfighters or as a team of intelligent autonomous systems. The tactical scenarios will serve to demonstrate the utility of metacognition, including reverse XAI and trust and machine-to-machine XAI and trust.

Metacognitive AI systems are poised to revolutionize tactical battle management – especially in complex military operations.²¹ Developing decision aids and autonomous systems that are synthetically metacognitive will be a game-changer for future warfare. It will lift much of the “fog of war” in terms of heightened situation awareness, especially in complex situations. It will leverage the intelligence of future decision aids and autonomous systems to optimize tactical courses of action that incorporate assessments of uncertainty and unknowns, and self-adapting goals based on how situations unfold. The primary end user is the tactical warfighter who must make critical military decisions in the midst of highly complex problem domains. Beyond just warfare, the achievement of synthetic metacognition is an important milestone in the advancement towards artificial general intelligence.²²

In conclusion, metacognition in AI is still evolving but holds great promise for creating more autonomous, self-aware, and adaptable systems. Research continues to advance in areas like meta-learning, uncertainty estimation, and self-reflection, which could ultimately lead to more robust AI systems capable of not only learning but also monitoring, adjusting, and improving their own processes.

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Dr. Bonnie Johnson is a professor of Systems Engineering at the Naval Postgraduate School. She leads interdisciplinary research for the Navy and DoD on artificial intelligence for tactical decision-making. She studies complexity theory, autonomous systems, weapon systems, directed energy systems and advanced warfare technologies. Dr. Johnson received her PhD in systems engineering from NPS, her M.S. in systems engineering from Johns Hopkins, and her B.S. in physics from Virginia Tech.